Workforce Strain in the Production of Accounting Estimates: Evidence from CECL Adoption

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Abstract

I examine whether an information-intensive accounting standard strains the specialized employees responsible for its implementation. Using regulatory filings and Glassdoor reviews classified by job function, I study U.S. banks' adoption of the current expected credit loss (CECL) standard through within-bank comparisons across employee functions. Following adoption, risk-function employees, who perform the forecasting and modeling required by CECL, report a decline in work-life balance relative to other employees in the same bank. Written reviews corroborate this increase in strain, with more complaints about workload and managerial support among risk-function employees. Credit-loss allowances are more predictive of subsequent net charge-offs when prior risk-function work-life balance is better relative to that of other employees. This association between workforce conditions and allowance informativeness is stronger under CECL than under the incurred-loss model, with the clearest statistical evidence at the two-quarter horizon. The findings document workforce strain as an employee burden of accounting regulation and connect it with the informativeness of accounting estimates.

JEL classification: G21; G28; J28; M41

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1. Introduction

Over the past two decades, accounting standards have increasingly relied on forward-looking, model-based, and judgment-intensive estimates in financial reporting[1]. These estimates can make reporting more relevant and timely (Beatty and Liao, 2014; Kim, Kim, Kleymenova and Li, 2026), but producing the required information can be costly. Their informational benefits must therefore be considered alongside the costs of implementation. As Ball (2024) emphasizes, however, many firm-level costs of accounting regimes are difficult to observe, and evidence on costs remains limited relative to evidence on benefits. Understanding these costs requires looking inside firms at the work involved in implementing new reporting requirements.

Accounting rules do not apply by themselves. They are implemented through people who supply the data, analysis, and judgment that complex estimates require. When a new standard requires firms to collect additional information or perform more extensive analysis, it can increase demands on the employees responsible for this work. Firms can accommodate these demands by hiring, reorganizing, or outsourcing. But if these adjustments are insufficient, the additional work can place pressure on employees' time and attention, potentially affecting both their working conditions and the quality of their output (Khavis and Krishnan, 2021). The potential consequences for output quality are especially relevant for forward-looking estimates, which rely on uncertain inputs that are often difficult to verify (Watts, 2003; Bonsall, Schmidt and Xie, 2025). Workforce conditions may therefore connect the burden of implementing a standard with the informational benefits it is intended to deliver.

This paper asks whether an information-intensive accounting standard strains the specialized workforce responsible for its implementation. I next examine how any strain manifests in employees' work environment. Finally, I ask whether the working conditions of that

[1] Examples include fair value measurement under ASC 820 and IFRS 13, variable consideration under ASC 606 and IFRS 15, and expected credit loss estimation under ASC 326 and IFRS 9.

workforce are associated with the informativeness of the resulting accounting estimate.

I examine these questions in the setting of U.S. banks' adoption of the current expected credit loss (CECL) standard. CECL requires banks to estimate lifetime expected credit losses using historical experience, current conditions, and reasonable and supportable forecasts (FASB, 2016). The setting has three features that make the questions tractable. First, the credit-risk assessment, forecasting, and validation work involved in producing these estimates draws directly on risk-function employees. Their identifiable roles permit a within-bank comparison of employees directly exposed to these demands with other employees of the same institution. Second, SEC filers other than smaller reporting companies generally adopted in 2020, while smaller reporting companies and other institutions generally adopted in 2023 following the FASB's deferral in ASU 2019-10 (FASB, 2019).[2] This timing allows the analysis to examine adoption in different calendar periods. Third, the informativeness of the standard's central output, the allowance for credit losses, can be assessed against subsequently realized credit losses (Altamuro and Beatty, 2010; Beatty and Liao, 2014).

I use work-life balance (*WLB*) ratings from bank employees' Glassdoor reviews as a proxy for workforce strain. Using self-reported job titles to identify risk-function reviews (Huang, Li and Markov, 2020), I construct *WLB_Gap* as their average *WLB* rating minus that of the remaining reviews from the same bank over the same review window[3]. The comparison nets out bank-level conditions that affect both groups equally, while bank fixed effects absorb persistent differences in their *WLB* gap. To characterize the strain captured by these ratings, I examine employees' written descriptions of workplace problems in the *Cons* section of each review. I classify these descriptions into distinct categories of workplace complaints and examine which become more prevalent among risk-function employees relative to the

[2] Adoption was not confined to these two years. Early adoption was permitted, and optional temporary relief under the CARES Act allowed some institutions in the earlier cohort to defer adoption (FASB, 2016; U.S. Congress, 2020).

[3] The comparison group includes some reviews without reported job titles. To assess sensitivity to their inclusion, I repeat the three primary analyses using workforce measures constructed solely from reviews with reported job titles. The results (reported in Section 4.4) are qualitatively similar to those from the main analyses.

comparison group following CECL adoption. To address the third research question, I assess allowance informativeness through the association between current allowances and subsequent net charge-offs. Comparing the allowance–*WLB_Gap* interaction across the incurred-loss and CECL regimes tests whether workforce conditions are more strongly associated with allowance informativeness under CECL.

I combine FR Y-9C regulatory filings for U.S. bank holding companies (BHCs) from 2015 through 2024 with employee reviews from Glassdoor.[4] Because Glassdoor coverage is concentrated among larger institutions, I collect reviews for BHCs reaching $5 billion in total assets during this period. Matching these BHCs and their subsidiary commercial banks to their Glassdoor pages yields more than 379,000 reviews. The resulting data support comparisons of working conditions across employee functions within the same bank. For example, the baseline *WLB* analysis draws on 1,960 bank-quarter observations across 128 BHCs with sufficient review coverage for both the risk function and the comparison group.

Using a difference-in-differences design around staggered CECL adoption, I find that risk-function *WLB* declines relative to the comparison group by 0.168 points, equivalent to roughly two-thirds of the sample mean of *WLB_Gap*. Decomposing the gap attributes this decline to deteriorating risk-function conditions, while the comparison group remains essentially unchanged. The gap remains depressed through the third post-adoption year, suggesting that the burden extends beyond the initial implementation period. Banks adopting in 2022 or later exhibit an even larger decline, reducing the concern that acute pandemic disruption drives the baseline result. The decline also persists across alternative estimation and inference approaches.

Employees' written reviews clarify the nature of this deterioration. Following adoption, workload and managerial support complaints increase among risk-function employees relative to the comparison group. These patterns point to heavier work demands and perceived

[4] Regulatory data through 2025Q4 are used solely to measure subsequent net charge-offs for observations near the end of the sample.

shortcomings in the support available to meet them. Together with the statistically insignificant estimates for the other Glassdoor rating gaps, the textual evidence weighs against generalized employee dissatisfaction as the alternative explanation for the decline in *WLB*.

I further find that better relative risk-function working conditions are more strongly associated with allowance informativeness under CECL than under the incurred-loss model. The cross-regime difference emerges at the two-quarter and four-quarter future-loss horizons, with no statistically detectable difference at one quarter. Among banks observed under both accounting regimes, a one-standard-deviation higher *WLB_Gap* corresponds to an increase in the four-quarter predictive slope of approximately 4.6 cents of subsequent net charge-offs per dollar of allowance under CECL, compared with less than one cent before adoption. The two-quarter cross-regime difference remains significant under wild-cluster bootstrap inference. The four-quarter estimates show a similar economic pattern, although the cross-regime difference is not significant under this inference procedure. Decomposing the workforce measure locates the positive change in the risk-function component, weighing against an explanation based on generally better bank-wide working conditions.

This study contributes to research on the economic consequences of accounting standards by documenting the implementation burden borne by employees. Prior research shows that new standards alter firms' information production and induce investments in personnel and systems (Shroff, 2017; Roh, 2025; Kim et al., 2026). Related studies document increased demand for accounting personnel and changes in the content and appeal of accounting work (Huang, Enache, Moldovan and Srivastava, 2025; Le, 2026). I show that CECL adoption is followed by a sustained deterioration in risk-function working conditions, with increased complaints about workload and managerial support. This evidence on the burden borne by employees responds to the measurement challenges and need for research on accounting costs highlighted by Ball (2024).

The study also contributes to research on human capital and financial reporting quality.

Prior studies link reporting quality to employee expertise and incentives (Armstrong, Kepler, Larcker and Shi, 2024; Barrios, Choi, Deller, Pacelli and Packard, 2026). Research on credit-loss estimation similarly highlights the role of borrower information and managerial characteristics (Yang, 2022; Bischof and Rudolf, 2025). I examine the working conditions of the employees involved in producing these estimates and show that their association with allowance informativeness is substantially stronger under CECL than under the incurred-loss model. This evidence points to the information requirements of accounting standards as an important consideration in understanding the relation between human capital and reporting quality.

These findings also inform standard setting and post-implementation review. Assessments of implementation can benefit from considering employees' working conditions alongside investments in personnel and systems, because those conditions reflect the burden of performing the required work and are associated with the informativeness of the resulting estimates.

The remainder of this study is organized as follows. Section 2 develops the hypotheses. Section 3 describes the data and sample. Section 4 presents the empirical results. Section 5 concludes.

2. Background and Hypothesis Development

Information-intensive accounting standards can increase the work required to produce financial reports by expanding the data, analysis, and judgment underlying accounting estimates. Meeting these requirements depends on the employees who perform the work and the resources available to them. Although many implementation costs are difficult to observe (Ball, 2024), recent studies document increased demand for accounting personnel during the implementation of new standards (Huang et al., 2025; Roh, 2025; Kim et al., 2026). This evidence indicates that firms seek additional resources, but does not establish whether those resources are sufficient to absorb the work required of employees.

CECL provides a setting in which these additional requirements can be traced to specific estimation activities. For financial assets measured at amortized cost, the standard replaces incurred-loss recognition with estimates of losses expected over their remaining contractual lives. Banks must incorporate historical experience, current conditions, and reasonable and supportable forecasts into these estimates (FASB, 2016). Producing this information places additional responsibilities on preparers. The FASB acknowledged this consequence when it noted that CECL would shift some loss-estimation costs from financial statement users to preparers, even as it expected costs for the system as a whole to decline (FASB, 2016, para. BC12).

These responsibilities involve substantial work in developing and maintaining the estimation process. In the FASB's post-implementation review, financial institutions reported considerable effort devoted to validating data, modifying business processes, and evaluating CECL models through parallel runs and backtesting (FASB, 2026). These activities draw on expertise in credit assessment, forecasting, and model validation. Consistent with the relevance of this expertise, Kim et al. (2026) identify credit-risk analytics and quantitative positions among the common titles in CECL-related job postings. Although implementation also involves other functions, risk-function employees are particularly exposed because their core responsibilities overlap with these estimation activities.

This exposure extends beyond establishing the initial estimation process. Banks must update their allowances at subsequent reporting dates, requiring them to reassess the information and judgments underlying their estimates as borrower conditions and economic expectations change (FASB, 2016). CECL therefore introduces recurring demands on the employees involved in producing these estimates. The potential burden depends on how banks accommodate this continuing work alongside employees' existing responsibilities.

Banks can accommodate the additional work through adjustments in staffing and work processes. They can recruit employees, redistribute responsibilities, invest in information systems, or obtain external assistance. The CECL-related job postings documented by Kim et al. (2026) provide evidence of efforts to obtain relevant expertise. These responses can expand the resources available to employees or reduce the effort needed to complete particular tasks. However, accommodating the work through these channels also entails costs and frictions. Recruiting and retaining specialists requires continuing personnel expenditures, while integrating new employees and implementing systems requires time and input from existing staff. Therefore, whether the demand shock is fully absorbed without increasing employee strain remains an empirical question. This discussion leads to the first hypothesis:

H1. Following CECL adoption, risk-function employees experience a greater increase in workload strain than other employees in the same bank.

The working conditions of risk-function employees may also matter for the information conveyed by the allowance for credit losses. Allowance informativeness concerns how well the estimate reflects information about subsequent credit losses. Producing an informative allowance requires banks to identify relevant borrower information, assess its implications, and incorporate those assessments into the reported estimate. The quality of this process matters. Altamuro and Beatty (2010) document improvements in loan loss provision validity following stronger internal control requirements, suggesting that credit-loss reporting depends on the process through which estimates are produced.

Employees' time and attention are important inputs into that process. Competing demands may limit their ability to verify data, reassess assumptions, and investigate unusual model outputs. Relevant changes in borrowers' repayment prospects may consequently receive insufficient attention or be inadequately reflected in the estimate. Related evidence links limited firm resources to internal control weaknesses and control deficiencies to lower accrual quality (Doyle, Ge and McVay, 2007; Ashbaugh-Skaife, Collins, Kinney Jr and La-

Fond, 2008). In the audit setting, Khavis and Krishnan (2021) find that better employee work-life balance is associated with higher audit quality. These findings support the possibility that the conditions under which employees perform accounting work are relevant to the quality of its output.

The importance of these conditions may differ across accounting regimes because the regimes differ in how employees' judgments enter the allowance. Under the incurred-loss model, recognition depended on evidence that losses were probable and had been incurred. Although this assessment required judgment, the recognition threshold limited the extent to which expectations about future losses could enter the reported estimate. CECL removes that threshold and requires banks to incorporate reasonable and supportable forecasts into lifetime expected-loss estimates (FASB, 2016). By making a broader set of expectations relevant to recognition, CECL increases the scope for employees' evaluation of uncertain information to influence the allowance.

Evidence on forecasting inputs supports this distinction. Canals-Cerdá (2026) shows that CECL estimates are more sensitive to macroeconomic forecast error and model misspecification than estimates under the incurred-loss framework. Using confidential supervisory forecasts, Bhojraj, Kleymenova, Liu, Lu and Sengupta (2025) document systematic forecast errors whose association with loan loss provisions is more pronounced among CECL adopters. Similarly, Bonaldi, Liang and Yang (2023) report pandemic-period patterns in provisioning and earnings quality that they interpret as greater accounting noise and reporting bias under CECL. These findings suggest that imperfections in forward-looking inputs can have important reporting consequences, making the conditions under which employees evaluate those inputs particularly relevant.

Banks may nevertheless protect the quality of allowance estimates from pressures on employees. They can prioritize estimation when allocating staff and managerial attention, preserving critical tasks while employees absorb additional demands elsewhere. Standardized systems and external specialists may reduce the effort required of individual employees, while review procedures and supervisory scrutiny may help identify shortcomings in the estimation process. These arrangements could allow banks to maintain estimate quality even when the employees involved experience greater strain.

The tension is therefore between CECL's greater reliance on employees' assessment of uncertain information and banks' ability to protect the estimation process from unfavorable working conditions. Whether the association between workforce conditions and allowance informativeness strengthens under CECL remains an empirical question. This discussion leads to the second hypothesis:

H2. More favorable risk-function working conditions are associated with greater allowance informativeness, and this association is stronger under CECL than under the incurred-loss regime.

3. Data and Sample

3.1. Sample Construction

I combine quarterly FR Y-9C regulatory filings for U.S. bank holding companies (BHCs) with employee reviews from Glassdoor. The focal analysis covers 2015Q1 through 2024Q4. Regulatory filings through 2025Q4 provide subsequent net charge-offs for observations near the end of this period. The main analyses are conducted at the BHC-quarter level.

I collect Glassdoor review histories for BHCs whose total consolidated assets reach at least $5 billion during the focal period. This restriction concentrates data collection on institutions likely to have sufficient reviews for comparisons across employee functions. I use the WRDS Bank Structure database to identify commercial-bank subsidiaries and their ultimate parent BHCs. Matching these institutions to Glassdoor requires reconciling legal entity names in regulatory records with the employer names used on the platform. Reviews may appear on pages associated with either the parent or its subsidiaries. I generate name variants, identify candidate pages, and manually review unresolved matches. Appendix C describes the matching procedure and data checks.

Each analysis uses a sample defined by its respective data requirements. The WLB and allowance informativeness analyses each require at least three total reviews from both the risk and comparison groups, with a non-missing *WLB* mean for each group within the relevant review window. The textual analysis requires at least three successfully classified *Cons* reviews from each group and does not impose a WLB-rating requirement.

The final WLB and textual estimation samples contain 1,960 and 1,966 BHC-quarter observations, respectively, each spanning 128 BHCs. The allowance informativeness tests apply additional requirements for historical review coverage within the same accounting regime[5] and the availability of subsequent credit-loss data. At the four-quarter future-loss horizon, the all-eligible sample contains 1,430 observations across 108 adopting BHCs. Restricting this sample to BHCs with usable observations in both regimes yields a same-bank sample of 970 observations across 45 BHCs. The review windows and remaining data requirements are described below and in the corresponding research-design sections.

3.2. Measures of Workforce Conditions

3.2.1. Employee reviews and work-life balance

Glassdoor allows current and former employees to evaluate their employers anonymously. Reviews provide an overall employer rating and can include separate ratings for Work/Life Balance, Senior Management, Compensation and Benefits, Career Opportunities, and Cul-

[5]The historical *WLB* window covers the four quarters preceding the allowance quarter. Requiring the review window and the allowance quarter to fall within the same accounting regime excludes the adoption quarter and the following three quarters.

ture and Values. These ratings range from one to five, with higher values indicating more favorable assessments. I refer to the Senior Management rating as Leadership in the tables. Reviews also contain posting dates, employment status, self-reported job titles when available, and written comments about the employer. The baseline construction retains reviews across employment statuses and locations.

I use the Work/Life Balance (*WLB*) rating to capture workload strain reflected in employees' ability to balance work with life outside work. Greater demands on employees' time and attention can make that balance more difficult to maintain. The *WLB* rating therefore provides a more direct measure of this aspect of strain than the overall employer rating, which combines assessments of multiple aspects of employment. Prior research also uses Glassdoor *WLB* ratings to study working conditions in accounting firms (Khavis and Krishnan, 2021). The measure captures employees' reported experiences rather than directly observing hours worked or staffing capacity. I also examine the other rating dimensions as comparison outcomes in Section 4.1.2.

I complement WLB ratings with employees' written descriptions in the *Cons* section of their reviews. These descriptions provide more granular accounts of workplace experiences and allow me to assess whether the textual evidence corroborates the patterns in *WLB*. The classification procedure and related measurement checks are described in Section 4.2.1.

3.2.2. Function classification

I identify employee functions using reviewers' self-reported job titles, building on prior research that extracts function-specific information from employee reviews (Huang et al., 2020). I standardize capitalization, spacing, and punctuation and remove designated modifiers from the beginning and end of each title. The resulting base roles are matched to a predefined classification dictionary. Titles absent from the dictionary are assigned using keyword rules. Appendix C describes the classification procedure.

The classification distinguishes the risk and economics group from other functions. The risk group includes credit analysts, underwriters, risk managers, model validation, and quantitative analysts, along with roles in economics, actuarial work, portfolio management, and investment research. The comparison group combines all remaining reviews associated with the same BHC, including reviews without reported job titles. Titles that remain unmatched are also assigned to this group. Section 4.4 examines the sensitivity of the results to excluding reviews without reported titles.

3.2.3. The WLB_gap

I aggregate individual ratings to measure risk-function work-life balance relative to the comparison group. For H1, the review pool for BHC i in quarter t includes reviews posted during quarters t through $t+3$. Pooling four quarters provides more observations for the function-level averages, although it reduces the precision with which changes can be dated.[6] Within each group, I average all nonmissing WLB ratings in this window, giving equal weight to each observed rating. The measure of differential operating conditions is defined for BHC i in quarter t as

$$\text{WLB\_gap}_{i,t} \;=\; \overline{\text{WLB}}^{\text{risk}}_{i,t} \;-\; \overline{\text{WLB}}^{\text{non-risk}}_{i,t}. \tag{1}$$

A higher value indicates more favorable risk-function work-life balance relative to the comparison group. Changes in the component means indicate how each group contributes to changes in the gap.

Comparing the two groups within the same bank and review window removes influences that affect both groups' ratings equally. The gap remains sensitive to changes in reviewer composition and contemporaneous shocks that affect the groups differently. These considerations inform the research design and subsequent analyses.

[6]For example, the observation indexed to 2020Q1 uses reviews posted from January 1 through December 31, 2020.

3.3. Allowance Informativeness

For H2, I assess allowance informativeness through the association between the reported allowance and subsequently realized credit losses. An allowance that incorporates information about future losses should help explain subsequent net charge-offs. This approach draws on research evaluating credit-loss reporting using its relationship with subsequent losses (Altamuro and Beatty, 2010; Beatty and Liao, 2014).

I obtain allowances for loan and lease losses and net charge-offs from FR Y-9C filings. I measure the allowance at the end of quarter t and future losses as cumulative net charge-offs over the subsequent one, two, or four quarters. Both measures are scaled by gross loans and leases at the end of quarter t.

To relate allowance informativeness to prior workforce conditions, I construct a trailing *WLB_Gap* using reviews posted during quarters $t-4$ through $t-1$, applying the same function classifications and aggregation method as in H1. This window measures working conditions before the quarter-t allowance measurement. I retain observations only when quarter t and the preceding four quarters are either all before or all after CECL adoption, so that historical WLB and the current allowance correspond to the same accounting regime. Section 4.3.1 describes the regression specifications and estimation samples.

3.4. CECL Adoption

I identify CECL adoption using three fields in the FR Y-9C, Notes to the Income Statement—Other. BHCKJJ26 reports the cumulative adjustment to retained earnings upon adoption, BHCKJJ27 reports initial allowances recognized upon the acquisition of purchased credit-deteriorated assets, and BHCKJJ28 reports the initial adoption effect on allowances for loans and leases held for investment and held-to-maturity debt securities (Board of Governors of the Federal Reserve System, 2020). I define the adoption quarter, τ_i, as the first quarter in which any of these fields is nonmissing. The indicator $Post_{i,t}$ equals one when $t \geq \tau_i$ and zero otherwise.

Adoption is concentrated in two principal cohorts. Under ASU 2019-10, SEC filers other than smaller reporting companies were generally required to adopt for fiscal years beginning after December 15, 2019, while other entities were required to adopt for fiscal years beginning after December 15, 2022 (FASB, 2019). These effective dates generally correspond to 2020 and 2023 adoption for calendar-year institutions. Early adoption and temporary relief under the CARES Act also generated adoption outside these principal cohorts (FASB, 2016; U.S. Congress, 2020). I classify institutions adopting in 2022Q1 or later as late adopters. Section 4.1.2 examines this group separately to assess adoption occurring after the initial pandemic disruption.

3.5. Descriptive Statistics

Table 1 reports summary statistics for the WLB and allowance informativeness analyses. Panel A describes the *H1* estimation sample of 1,960 BHC-quarter observations across 128 BHCs. The mean *WLB_gap* is 0.25 with a standard deviation of 0.63. Average ratings are 3.76 for risk-function WLB and 3.51 for non-risk WLB, indicating more favorable reported work-life balance among risk-function reviewers within the same bank and review window, on average. The gap ranges from −0.45 at the 10th percentile to 1.00 at the 90th percentile.

The four-quarter review windows contain substantially fewer risk-function reviews than non-risk reviews, with a mean (median) of 21.6 (7) and 590 (139), respectively. Approximately two-thirds of the BHC-quarter observations fall in the post-adoption period. Mean log assets are 18.13, and the mean capital ratio is 11.3 percent.

Panel B describes the same-bank sample used in the *H2* allowance informativeness tests at the four-quarter horizon. It contains 970 BHC-quarter observations across 45 BHCs with usable observations in both accounting regimes. The allowance averages 1.52 percent of gross loans, while cumulative four-quarter net charge-offs average 0.56 percent, with a 90th percentile of 2.00 percent. The trailing *WLB_gap* has a mean of 0.24 and a standard deviation of 0.51. The sample in Panel B comprises larger banks on average, with mean log assets of 18.76 compared with 18.13 in Panel A.

4. Research Design and Results

4.1. Workforce Strain Following CECL Adoption

This section tests *H1.* Section 4.1.1 presents the baseline difference-in-differences estimates, dynamic pattern around adoption, and robustness tests of the results. The subsequent subsection addresses alternative explanations regarding general employee dissatisfaction and pandemic-period disruption.

4.1.1. Does CECL Strain Risk-Function Employees?

The baseline specification is

$$\text{WLB\_Gap}_{i,t} \;=\; \beta\, Post_{i,t} \;+\; \Gamma' X_{i,t} \;+\; \alpha_i \;+\; \gamma_t \;+\; \varepsilon_{i,t}, \tag{2}$$

where i and t index BHC and calendar quarter, respectively. The dependent variable, $WLB\_Gap_{i,t}$, is the mean work-life balance rating of risk-function reviews minus the mean rating of non-risk reviews, as defined in Section 3.2.3. The variable of interest is $Post_{i,t}$, an indicator equal to one in and after the quarter in which BHC i adopts CECL and zero otherwise. The coefficient of interest, β, captures the average within-bank change in the risk-minus-non-risk work-life balance gap following CECL adoption, relative to contemporaneous changes at other banks in the sample. If CECL's incremental estimation demands fall disproportionately on the risk function, as *H1* predicts, β would be negative. The control vector $X_{i,t}$ includes log assets, the capital ratio, the earnings before loan loss provisions ratio, and the deposit ratio, as defined in Appendix A. BHC fixed effects, α_i, absorb persistent differences across banks in the WLB gap, while calendar-quarter fixed effects, γ_t, absorb common changes over time. Identification relies on the assumption that, absent CECL adoption, the WLB gap would have followed parallel trends across banks, conditional on the controls. Standard errors are clustered by BHC to account for within-bank serial correlation, including that induced by overlapping review windows.

Column (1) of Table 2 reports the baseline estimate. Consistent with *H1*, the coefficient on *Post* is negative and statistically significant (-0.168, $p < 0.05$). Risk-function WLB declines relative to non-risk WLB within the same bank following CECL adoption. The estimated decline of 0.168 points is equivalent to approximately two-thirds of the full-sample mean *WLB_gap*.

Figure 1 examines the timing of this decline by replacing $Post_{i,t}$ in Equation (2) with event-time indicators from $k = -8$ through $k = 12$. The endpoint indicators combine observations at $k \leq -8$ and $k \geq 12$, respectively. Because the WLB measure for quarter t pools reviews posted during quarters t through $t + 3$, observations dated $k = -3$ through $k = -1$ include post-adoption reviews. I therefore use $k = -4$, the last window containing only pre-adoption reviews, as the reference period and distinguish the overlapping windows in the figure.

The coefficients for windows containing only pre-adoption reviews are small, with absolute magnitudes below 0.04, and jointly insignificant ($p = 0.974$). These estimates show no systematic differential trend before adoption. Within the overlapping window, the coefficient turns negative at $k = -1$, when the review pool already includes the adoption quarter and the following two quarters.

All thirteen post-adoption coefficients are negative, ranging from approximately -0.10 to -0.33. The estimates at $k = 2$ and $k = 3$ are statistically significant at the five percent level. The consistently negative estimates at longer horizons suggest that the deterioration in relative risk-function WLB extends beyond the initial transition.

The negative gap estimate could reflect a decline in risk-function *WLB*, an improvement in non-risk *WLB*, or both. To identify which component accounts for the result, I re-estimate Equation (2) separately for risk-function and non-risk WLB. Both regressions use the same observations, controls, and fixed effects as the baseline specification, so the gap coefficient equals the difference between the two component coefficients.

Columns (2) and (3) of Table 2 report the results. The coefficient on *Post* is negative for risk-function WLB (-0.168, $p = 0.068$) and close to zero for non-risk WLB (-0.0004, $p = 0.992$). The risk-function coefficient is nearly identical in magnitude to the baseline gap coefficient. The decomposition therefore indicates that the estimated deterioration is concentrated in risk-function ratings, consistent with *H1*.

I assess whether individual BHCs disproportionately influence the baseline estimate. Figure 2 addresses this concern by re-estimating Equation (2) 128 times, omitting one BHC per iteration, and plotting the resulting coefficients in sorted order. Every iteration produces a negative estimate, with coefficients ranging from -0.199 to -0.133 around the full-sample estimate of -0.168. All 128 estimates are significant at the ten percent level, and 88 percent are significant at the five percent level. Overall, the leave-one-out test estimates are stable in sign and magnitude to the exclusion of any individual BHC.

Another feature of the setting, the staggered adoption structure, raises a separate estimation concern. When treatment effects vary over time, two-way fixed effects estimates place weight on comparisons in which banks treated earlier serve as controls for banks treated later, and these comparisons can bias the estimate toward zero or reverse its sign (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). Appendix Table B2 reports estimates from the Callaway and Sant'Anna (2021) estimator, which restricts comparisons to not-yet-treated and never-treated banks and, in this setting, shifts each bank's effective adoption date three quarters earlier so that no measurement containing post-adoption reviews enters a control group. The average treatment effect across event quarters zero through four is -0.287 ($p = 0.026$), consistent in sign and significance with the baseline estimate.

4.1.2. Is the Decline Attributable to CECL?

Two alternative explanations could account for the baseline estimate. First, risk-function employees may have become generally dissatisfied around adoption for reasons unrelated to workload, such as compensation decisions, organizational disturbance, or the standing of the function within the bank. Under this explanation, the gap decline reflects discontent rather than strain. Second, the first adoption cohort implemented CECL in the opening quarter of the COVID-19 pandemic, a period in which loan workouts, forbearance processing, and pandemic-related stress modeling plausibly raised risk-function workload independently of the standard. Under this explanation, the estimate reflects pandemic disruption rather than CECL. Each explanation implies a distinct empirical prediction, and I test each in turn.

If risk-function employees became generally dissatisfied, the deterioration should appear across the dimensions of the employment experience rather than concentrating in work-life balance. Following Khavis and Krishnan (2021), who examine the remaining Glassdoor rating dimensions alongside work-life balance, I construct risk-minus-non-risk gaps for the overall rating and the four sub-ratings (career-opportunities, compensation-and-benefits, culture-and-values, and senior-leadership) and estimate Equation (2) for each dimension. Table 3 reports the results. The coefficient on *Post* is statistically insignificant for every alternative dimension, with estimates ranging from -0.079 to 0.032, compared with -0.168 ($p < 0.05$) for work-life balance. The average coefficient across the five placebo dimensions is -0.024, and the difference between the *WLB* coefficient and the average placebo coefficient is -0.144 ($p = 0.094$). The larger decline in WLB relative to the other rating dimensions is consistent with increased workload strain and weighs against generalized employee dissatisfaction.

The compensation-and-benefits dimension warrants separate interpretation, because its expected response to a workload shock differs from that of the other dimensions. Employees evaluate compensation relative to both outside options and the effort their jobs require. Greater demands can lower these ratings if pay remains unchanged, but compensating pay adjustments may offset this response. Such adjustments are consistent with the increase in salary and employee benefit expenses following CECL adoption documented by Kim et al. (2026). The insignificant compensation-gap estimate therefore does not imply that pay was unchanged. Higher pay can compensate employees for additional work without reducing its demands on their time. Compensation ratings can consequently remain stable even as work-life balance deteriorates.

Later adoption provides a separate test of whether the baseline result is specific to the coincidence of CECL implementation and the pandemic's onset. If pandemic-period disruption rather than CECL produced the baseline estimate, banks adopting well after the acute pandemic period should exhibit little or no gap decline. Panel B of Table 2 tests this prediction by estimating Equation (2) on the late-adopter cohort, excluding early adopters who adopt CECL before 2022.

The results are inconsistent with the alternative explanation. The coefficient on *Post* in column (4) is negative and statistically significant (-0.606, $p < 0.01$). Its magnitude is approximately 96 percent of the *WLB_gap*'s standard deviation in the full *H1* sample. Columns (5) and (6) again locate most of the estimated decline in the risk-function component (-0.543, $p < 0.05$), while the non-risk coefficient is positive and statistically insignificant (0.063). The larger effect among late adopters is consistent with the mechanism underlying *H1*. Late adopters are smaller institutions with thinner risk-management infrastructure, and the same implementation demands spread across fewer specialized employees imply a heavier burden per risk-function employee.

Given the 35 BHC clusters, I also assess inference using a wild-cluster bootstrap with

Webb weights and 9,999 replications. As reported in the notes to Table 2, the late-adopter gap estimate remains significant ($p = 0.025$). In a leave-one-bank-out analysis, all 35 estimates remain negative and significant at the five percent level, ranging from -0.708 to -0.468. The relative deterioration in risk-function WLB therefore also appears among banks adopting after the initial pandemic disruption and persists when any single bank is excluded.

4.2. The Nature of the Strain Underlying the WLB Deterioration

A deterioration in perceived work-life balance may reflect many workplace problems, including pure workload increase, weaker managerial support, frictions in completing or coordinating work, dissatisfaction with compensation or career opportunities, or other features of the workplace. Thus, while the results in Section 4.1 establish that CECL adoption is associated with a deterioration in risk-function employees' *WLB*, the rating itself provides limited evidence on the nature of the underlying strain. To better understand what this deterioration reflects, I examine the employee-written *CONS* text in Glassdoor reviews and classify the complaints employees report into distinct workplace categories. This analysis provides a more granular view of how employees describe their work environment and allows me to examine which dimensions of that environment change following CECL adoption.

I proceed in two steps. Before examining whether particular complaints become more common following CECL adoption, Section 4.2.1 first evaluates what the textual complaint categories capture. Glassdoor reviews report not only employee-written comments but also several contemporaneous ratings of the employee's work experience. I therefore examine how the complaint categories relate to these ratings and whether the resulting associations align with the intended content of each category. Of particular interest is whether workload and time-demand complaints are more closely associated with *WLB* than with broader assessments such as overall satisfaction, career opportunities, compensation, culture, or leadership. This exercise helps establish that the textual measures capture distinct dimensions of employees' work experience rather than simply reflecting general dissatisfaction.

Section 4.2.2 then examines whether the incidence of these complaint categories changes following CECL adoption, separately for risk and non-risk employees. Combining the two analyses allows the textual evidence to speak more directly to the nature of the deterioration in *WLB* documented in Section 4.1.

4.2.1. Mapping Complaint Categories to WLB

To distinguish among the workplace problems described in employee reviews, I draw on the Job Demands–Resources (JD–R) framework, which organizes working conditions around job demands that require sustained employee effort and job resources that facilitate employees' ability to perform their work and cope with those demands (Demerouti, Bakker, Nachreiner and Schaufeli, 2001; Bakker and Demerouti, 2007). Related work-design research emphasizes that employees' work experience is multidimensional, distinguishing among characteristics such as workload, autonomy, social support, and features of the work context (Morgeson and Humphrey, 2006). Consistent with this view, prior research on occupational stress also treats quantitative workload and organizational constraints as distinct workplace stressors (Spector and Jex, 1998). Guided by these distinctions, I classify employee complaints into eight categories. On the demand side, the categories capture *Workload*, *Workflow Frictions*, *Role Pressure*, and *Job Security*. On the resource side, they capture deficiencies in *Autonomy*, *Managerial Support*, *Career Rewards*, and *Organizational Justice*. Appendix Table D1 provides the definitions and coding boundaries for each category.

I identify these workplace problems from the employee-reported *Cons* section of each Glassdoor review. I use GPT-5.6 Luna to read the review text and determine whether each of the eight complaint categories is present. The categories are coded independently, allowing a single review to identify multiple workplace problems. To keep the textual classification independent of the empirical setting, the model receives only the *Cons* text without any background information regarding the employer, employee job function, review date, employee ratings, or CECL adoption status. Reviews containing no meaningful negative complaint are classified as *No Complaint*. Reviews containing a meaningful negative complaint that does not fall into any of the eight categories are classified as *Other Complaints*. Additional classification details and the classification rules are provided in Appendix D.1.

Appendix Table D2 reports the distribution of the resulting classification. Among the 430,215 reviews with nonempty *Cons* text, 8.93% contain no substantive complaint and 10.80% contain a substantive complaint outside the eight categories. Thus, among reviews that contain a substantive complaint, approximately 88% fall into at least one of the eight complaint categories. These patterns indicate that the taxonomy captures a broad range of the workplace problems employees disclose while retaining a residual category for complaints that fall outside the theoretically motivated dimensions.

I next examine whether the complaint categories capture the workplace dimensions they are intended to measure. Following the spirit of Cai, Prat and Yu (2026), who validate a text-based measure by examining whether it exhibits empirical relations implied by the underlying construct, I test whether the complaint categories map into employees' contemporaneous Glassdoor ratings in predictable ways. If the categories distinguish among different dimensions of employees' work experience rather than simply capturing general dissatisfaction, their associations should differ across the structured ratings. Appendix Table D3 examines these associations among risk-function employees. Each column reports a review-level regression of one of the six Glassdoor ratings on all eight complaint categories jointly. The regressions use a common sample across rating outcomes and include Glassdoor-page and review-quarter fixed effects. Given the large sample and the natural association between negative review content and lower employee sentiment, statistical significance alone is not particularly informative in this setting. I therefore focus on differences in the economic magnitudes of the coefficients across rating dimensions.

The results are consistent with these predicted mappings. Workload complaints have the strongest association with *WLB*. Conditional on the rest of the complaint categories, a *workload* complaint is associated with a 0.935-point lower *WLB* rating on the five-point scale, compared with associations of 0.114 to 0.281 points in absolute value with the other five ratings. *Managerial Support* complaints exhibit a broader pattern: they are strongly associated with *WLB*, but their largest associations are with *Leadership*, *Culture*, and *Overall Rating*. *Career Rewards* complaints provide a particularly useful contrast. They have essentially no association with *WLB*, while their strongest associations are with *Career Opportunities* and *Compensation* ratings. *Organizational Justice* complaints similarly map more strongly into broader evaluations of *Culture*, *Leadership*, and *Overall Rating*, whereas *Workflow Friction* complaints have only a weak association with *WLB*. The remaining categories exhibit more moderate associations across rating dimensions. Taken together, the cross-rating patterns are consistent with the complaint categories capturing distinct aspects of employees' work experience rather than a common negative-sentiment factor.

One concern is that the particularly strong mapping between *Workload* complaints and *WLB* could arise mechanically because explicit complaints about work-life balance are themselves classified as workload complaints. Panel B of Appendix Table D3 therefore excludes reviews whose *Cons* text directly refers to work-life balance or WLB. The association between workload complaints and *WLB* remains economically large at -0.775, compared with associations ranging from -0.093 to -0.215 for the other five ratings. Thus, the close mapping between *Workload* complaints and *WLB* does not depend on reviews that explicitly repeat work-life-balance terminology. I next use these textual measures to examine which workplace problems change following CECL adoption.

4.2.2. *Changes in Employee Complaints Following CECL Adoption*

I next examine which types of employee complaints change following CECL adoption. For each complaint category, I construct three bank-quarter outcomes: the complaint rate among risk-function employees, the complaint rate among non-risk-function employees, and the difference between the two. The complaint rate is defined as the share of classified reviews assigned to the relevant category. I then re-estimate Equation 2 separately for each of these three outcomes and for each of the eight complaint categories. Table 4 therefore summarizes 24 separate regressions. All regressions use a common sample of 1,966 BHC-quarter observations across 128 BHCs, constructed independently of the WLB-rating sample. Each row reports the *Post* coefficients for one complaint category, with columns presenting the gap, risk-function, and non-risk-function estimates.

Workload and *Managerial Support* are the two categories with statistically significant increases in complaint rates among risk-function reviewers. Following CECL adoption, the risk-minus-non-risk gap in workload complaints increases by 4.0 percentage points. This change is driven by a 3.5-percentage-point increase in workload complaints among risk-function employees, with essentially no corresponding change among non-risk employees. The complaint gap for managerial support increases by 5.3 percentage points. Again, the change is concentrated among risk-function employees, whose managerial-support complaint rate increases by 4.4 percentage points, while the corresponding change among non-risk employees is small and statistically insignificant. These results help unpack the deterioration in *WLB* documented in Section 4.1.1. Following CECL adoption, risk-function employees become more likely to report complaints about both workload and managerial support, while neither complaint category increases among non-risk-function employees.

The concentration of the changes in *Workload* and *Managerial Support* is particularly informative in the CECL setting. The increase in *Workload* complaints captures the most direct dimension of the implementation burden: CECL requires greater forecasting, validation, documentation, and review effort from the employees involved in the estimation process. Consistent with this interpretation, the FASB's post-implementation review reports that banks devoted substantial time and effort to data validation, business-process changes, model validation, parallel runs, and backtesting during CECL implementation (FASB, 2026). These implementation demands also required firms to adapt existing processes and coordinate unfamiliar estimation activities.

The increase in *Managerial Support* complaints points to the resource side of this adjustment. Implementing a novel and judgment-intensive standard requires not only additional employee effort but also managerial guidance, coordination, feedback, and prioritization as employees adapt to unfamiliar models and processes. These managerial resources may themselves become strained during implementation, as managers must interpret the new requirements, coordinate changes across functions, and support employees while confronting the same transition. Thus, the joint increase in workload and managerial-support complaints is consistent with CECL simultaneously increasing the demands placed on risk-function employees and straining the organizational resources available to help them meet those demands.

The remaining categories show a different pattern. The *Career Rewards* complaint gap increases by 6.9 percentage points ($p < 0.05$), combining a statistically insignificant 3.5-percentage-point increase among risk-function reviewers with a 3.3-percentage-point decline among non-risk reviewers ($p < 0.01$). The gap coefficients for *Autonomy*, *Workflow Frictions*, *Role Pressure*, *Job Security*, and *Organizational Justice* are statistically insignificant.

The textual evidence corroborates the WLB findings by showing that risk-function reviewers report more complaints about workload and managerial support following CECL adoption. I next examine whether risk-function working conditions are associated with the informativeness of banks' credit-loss allowances.

4.3. Allowance Informativeness and Risk-Function Work Conditions

This section tests *H2*. Section 4.3.1 presents the design and examines how the relation between workforce conditions and allowance informativeness differs before and after CECL. Section 4.3.2 evaluates the sensitivity of the findings to alternative inference and the influence of individual banks.

4.3.1. Allowance Informativeness and Workforce Conditions Before and After CECL

I assess allowance informativeness through its association with subsequently realized credit losses. This approach draws on research evaluating credit-loss reporting through its relation to subsequent losses (Altamuro and Beatty, 2010; Beatty and Liao, 2014). I use the allowance because it directly reflects the estimated credit losses on the outstanding loan portfolio, whereas the provision also reflects the replenishment of charge-offs recognized during the period. Under CECL, the allowance incorporates expected losses over the remaining contractual lives of the loans (FASB, 2016). I examine subsequent net charge-offs over one, two, and four quarters, allowing progressively longer periods for losses to materialize.

For each future-loss horizon $h \in \{1, 2, 4\}$, I estimate,

$$\begin{aligned}\text{FutNCO}^h_{i,t} \;=\;& \beta_{1r}\,\text{Allowance}_{i,t} \;+\; \beta_{2r}\,\text{WLB\_Gap}^{tr}_{i,t} \;+\; \beta_{3r}\,\text{Allowance}_{i,t} \times \text{WLB\_Gap}^{tr}_{i,t} \\ &+\; \Gamma'_r X_{i,t} \;+\; \alpha_{i,r} \;+\; \gamma_{t,r} \;+\; \varepsilon_{i,t},\end{aligned} \tag{3}$$

where $r \in \{Pre, Post\}$ denotes the accounting regime. $\text{FutNCO}^h_{i,t}$ is net charge-offs accumulated over quarters $t+1$ through $t+h$, divided by gross loans at quarter t. $\text{Allowance}_{i,t}$ is the quarter-end allowance for loan and lease losses divided by quarter-end gross loans. These finite horizons capture subsequent loss realizations within the indicated periods rather than the complete realization of CECL's lifetime loss estimate. $\text{WLB\_Gap}^{tr}_{i,t}$ measures relative risk-function work-life balance using reviews posted during quarters $t-4$ through $t-1$.

This timing places workforce conditions before the allowance measurement. I also require quarter t and all four preceding quarters to fall within the same accounting regime, so that the workforce measure reflects conditions under the regime governing the allowance. Each group must have at least three total reviews and a nonmissing WLB mean over the historical window. The control vector $X_{i,t}$ includes bank size, capitalization, loan growth, the earnings-before-loan-loss-provisions ratio, the deposit ratio, current-quarter net charge-offs, and the current change in nonperforming loans. All regressors and BHC and calendar-quarter fixed effects are fully interacted with the accounting regime. This specification allows both slopes and fixed effects to differ before and after CECL while permitting direct tests of coefficient differences across regimes. Standard errors are clustered by BHC, allowing errors from the same bank to be correlated within and across regimes.

The coefficient of interest, β_{3r}, captures how the association between the allowance and subsequent losses varies with prior risk-function WLB relative to the comparison group. A positive coefficient indicates that the allowance is more strongly associated with future losses when relative risk-function working conditions are more favorable. *H2* predicts both a positive β_{3r} and a stronger dependence under CECL, represented by $\beta_{3,Post} - \beta_{3,Pre} > 0$.

I estimate the model using two samples of CECL adopters. The all-eligible sample includes every BHC-quarter satisfying the review, accounting-regime, and financial-data requirements for the relevant horizon. The same-bank sample additionally requires each BHC to have observations in both regimes in the final estimation sample for that horizon. The first sample provides broader coverage, while the second holds fixed the set of banks represented before and after CECL. Table 1, Panel B, describes the same-bank sample at the four-quarter horizon.

Table 5 reports the estimates. In the all-eligible sample, the one-quarter post-CECL interaction and its Post $-$ Pre difference are small and statistically insignificant. Positive associations emerge at the longer horizons. At two quarters, the post-CECL interaction is

0.0303 ($p < 0.01$), with a $Post - Pre$ difference of 0.0305 ($p < 0.05$). At four quarters, the corresponding estimates are 0.0895 and 0.0818, both significant at the one percent level. The pre-CECL interactions are statistically insignificant at all three horizons. Appendix Table B3 shows that the positive post-CECL interaction and cross-regime difference at the four-quarter horizon persist as controls are added sequentially.

The same-bank estimates retain this pattern. The one-quarter interaction and cross-regime difference remain insignificant. At two quarters, the post-CECL interaction is 0.0357 and the $Post - Pre$ difference is 0.0363, both significant at the one percent level. At four quarters, the post-CECL interaction is 0.0904 ($p < 0.01$), compared with an insignificant pre-CECL interaction of 0.0062. Their difference is 0.0842 ($p < 0.05$). The stronger association between workforce conditions and allowance informativeness under CECL therefore appears both in the broader sample and among banks represented in both regimes.

The magnitude is economically meaningful. In the four-quarter same-bank sample, the standard deviation of the trailing *WLB_gap* is approximately 0.51. A one-standard-deviation higher gap is therefore associated with a post-CECL allowance slope that is approximately $0.51 \times 0.0904 = 0.046$ larger. Holding gross loans fixed, this means that a one-dollar difference in the allowance predicts about 4.6 cents more in subsequent net charge-offs at the higher WLB gap. The corresponding pre-CECL slope difference is approximately $0.51 \times 0.0062 = 0.003$, or 0.3 cents per dollar of allowance.

Following the component analysis in Table 2, I examine the risk-function and non-risk-function WLB measures underlying the gap. In Table 2, the gap is the dependent variable, and separate regressions for its two components show which group's ratings account for its decline. Here, *WLB_gap* enters as an explanatory variable that moderates the association between the allowance and future losses. I therefore enter both *WLB_gap* components and their allowance interactions jointly, with regime-specific coefficients. This specification allows each function's working conditions to have a distinct association with allowance informativeness.

The samples, controls, and fixed effects remain the same as in Table 5.

The risk-function interaction increases after CECL in both samples. In the all-eligible sample, the $Post - Pre$ change is 0.0310 at two quarters ($p < 0.10$) and 0.0871 at four quarters ($p < 0.01$). The corresponding non-risk changes are negative and statistically insignificant. In the same-bank sample, the risk-function changes are 0.0388 ($p < 0.05$) and 0.0915 ($p < 0.01$), while the non-risk changes are -0.1034 and -0.2339, both significant at the ten percent level.

4.3.2. Does the Informativeness Pattern Withstand Scrutiny?

The same-bank estimates use 45 BHC clusters at the two- and four-quarter horizons. I examine whether inference is sensitive to this number of clusters and whether individual banks disproportionately influence the estimates. Table 7 reports wild-cluster bootstrap tests and leave-one-BHC-out diagnostics for these samples.

I supplement conventional BHC-clustered inference with a wild cluster bootstrap (Cameron, Gelbach and Miller, 2008), using Webb weights and 9,999 replications. The two-quarter results remain significant under this procedure. The post-CECL interaction has a bootstrap p-value of 0.0005, and the $Post - Pre$ difference has a bootstrap p-value of 0.0025. The two-quarter evidence that allowance informativeness depends more strongly on relative risk-function working conditions under CECL therefore remains significant under bootstrap inference.

At four quarters, the positive post-CECL interaction remains significant at the ten percent level under bootstrap inference ($p = 0.0985$). The $Post - Pre$ difference is positive and significant under conventional BHC-clustered inference ($p = 0.0249$), but not under the bootstrap ($p = 0.2000$). The four-quarter estimates therefore retain the economic pattern documented in Table 5, although statistical support for the cross-regime difference is sensitive to the inference procedure. The two-quarter horizon provides stronger statistical evidence that the relation strengthens under CECL. I next examine whether the positive estimates at both horizons are disproportionately influenced by individual BHCs.

To assess the influence of individual institutions, I re-estimate Equation (3) after excluding each of the 45 BHCs in turn. At two quarters, the post-CECL interaction ranges from 0.0241 to 0.0443, and the $Post - Pre$ difference ranges from 0.0252 to 0.0697. Both estimates remain positive and significant at the ten percent level in every iteration using conventional BHC-clustered inference. At four quarters, the corresponding ranges are 0.0599 to 0.1253 and 0.0540 to 0.1286. Both estimates remain positive in every iteration and significant at the ten percent level in 44 of the 45 iterations (97.8 percent). The point estimates are therefore stable to excluding individual banks, with the two-quarter results providing the clearest statistical support for *H2* across the inference procedures.

4.4. Job-Title Availability and Robustness

The comparisons between risk-function and other employees rely on reported job titles to identify employees involved in risk-related activities. In the baseline construction, reviews without reported titles remain in the non-risk benchmark. These reviews contain information about employees' working conditions, but their functional affiliation cannot be verified. I therefore examine whether including them materially changes the workforce measures and whether the main findings persist when these reviews are excluded.

Table 8, Panel A compares the two *WLB* constructions using the 1,955 bank-quarter observations included in both baseline *H1* estimation samples. Risk-function *WLB* is identical under the two constructions, since identifying risk-function employees already requires a reported title. The means and medians of non-risk *WLB* and the *WLB_gap* are also closely aligned. The mean *WLB_gap* is 0.2561 when reviews without titles are retained and 0.2550 when they are excluded. The correlations between the two constructions are 0.9621 for non-risk *WLB* and 0.9862 for the *WLB_gap*. In general, excluding reviews without titles produces little change in the average measured WLB differential and largely preserves its variation across bank-quarters in the *H1* sample.

Restricting the analysis to reviews with reported titles also changes review coverage unevenly over time. Table 8, Panel B reports title availability in the source review corpus for calendar years 2015 through 2025, before applying the estimation-sample restrictions. Of the 379,400 reviews posted during this period, 74,516, or 19.64 percent, lack a reported title. The proportion ranges from 45.85 to 55.76 percent during 2015 through 2018, compared with 5.60 to 12.96 percent during 2020 through 2025. A title requirement therefore removes a substantially larger share of early reviews, changing the composition and amount of information available to measure working conditions over time.

In this study, the comparison group serves to capture working conditions shared across employees within the same bank. Reviews without titles can contribute information about these conditions even though they do not establish the reviewer's function. Requiring a reported title provides a clearer basis for functional classification but disproportionately reduces early review coverage. Given this tradeoff and the close correspondence between the two *WLB* constructions, I retain the broader comparison group in the main analyses and examine the exclusion of reviews without titles as an alternative measurement approach.

Table 9 reports the three primary analyses of this study using the workforce measures constructed exclusively from reviews with reported job titles. Each analysis reapplies its original review-count requirements after excluding reviews without titles and retains the original review window, controls, fixed effects, and clustering specification. Panel A continues to show a decline in risk-function *WLB* relative to the comparison group following CECL adoption. The estimated decline in the *WLB_gap* is 0.156 points in the baseline sample and 0.590 points among late adopters, significant at the 10 percent and 1 percent levels, respectively. Panel B shows that the relative incidence of workload complaints and complaints about managerial support increases by 3.4 and 5.2 percentage points, respectively. These estimates are significant at the 10 percent and 5 percent levels. Although the baseline *WLB* and workload gap estimates fall from significance at the 5 percent level to the 10 percent level, the direction and substantive interpretation of the findings are preserved.

Panel C repeats the allowance informativeness tests in Table 5 using the alternative construction. The $Post - Pre$ difference in the allowance–WLB gap interaction remains positive and statistically significant at the two-quarter and four-quarter future-loss horizons, both among all eligible banks and in the same-bank sample. The corresponding differences at the one-quarter horizon remain statistically insignificant. Across the three analyses, the primary findings therefore remain qualitatively similar when reviews without reported job titles are excluded.

5. Conclusions

This paper examines whether an information-intensive accounting standard strains the specialized workforce responsible for its implementation, how that strain manifests in employees' work environment, and whether workforce conditions are associated with the informativeness of the resulting accounting estimate. Using CECL adoption as the setting, I combine employee reviews classified by job function with bank regulatory filings to connect employees' reported working conditions with the informativeness of banks' credit-loss allowances.

The first set of results documents a deterioration in risk-function work-life balance following CECL adoption. The estimated decline relative to other employees in the same bank is 0.168 rating points, equivalent to roughly two-thirds of the sample mean of the WLB_gap. The component estimates attribute this decline to risk-function WLB. The negative pattern extends beyond the initial implementation period and also appears among banks adopting after the acute pandemic period. Employee-written reviews corroborate these findings. Following adoption, risk-function reviewers report more complaints about workload and managerial support, providing a more granular account of the workplace problems accompanying the *WLB* deterioration.

The allowance informativeness tests show that the association between the allowance and subsequent credit losses is more sensitive to prior relative risk-function working conditions under CECL. The cross-regime difference is statistically insignificant over the first subsequent quarter but positive and significant over the two-quarter and four-quarter horizons. The two-quarter results remain significant under wild-cluster bootstrap inference, while statistical support for the four-quarter cross-regime difference is sensitive to the inference procedure. The positive estimates at both horizons are stable to the exclusion of individual BHCs. In joint models that include both functions' WLB measures, the allowance interaction with risk-function WLB also increases after CECL.

The findings connect the organizational consequences of accounting standards with the process through which accounting estimates are produced. Implementation demands are reflected in the working conditions of specialized employees, and those conditions are associated with the informational properties of their accounting output. The contrast between the incurred-loss and CECL regimes suggests that the importance of workforce conditions depends on the accounting task. Their association with allowance informativeness is stronger when the standard requires greater reliance on forward-looking information and judgment.

Longer post-adoption panels could clarify whether this stronger association persists as CECL implementation matures or weakens as banks and employees become more accustomed to its requirements. The present findings also suggest a role for workforce conditions in evaluating implementation. Post-implementation reviews may benefit from considering employee workloads and managerial support alongside expenditures on personnel and systems and the informational properties of reported estimates. Such evidence would help standard setters assess the organizational capacity needed to produce the information their standards require.

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Figures and Tables

Table 1: **Summary Statistics**

Panel A: H1 WLB Gap Sample

Variable	N	Mean	SD	p10	p50	p90
WLB gap	1,960	0.25	0.63	-0.45	0.27	1.00
WLB (risk function)	1,960	3.76	0.69	2.90	3.81	4.50
WLB (non-risk function)	1,960	3.51	0.38	3.04	3.53	3.96
Post	1,960	0.67	0.47	0	1	1
Log(Assets)	1,960	18.13	1.56	16.00	18.07	20.54
Capital ratio	1,960	0.11	0.03	0.08	0.11	0.15
EBLLP ratio	1,960	0.02	0.03	0.01	0.02	0.05
Deposit ratio	1,960	0.69	0.18	0.45	0.77	0.85
N (risk reviews)	1,960	21.56	37.10	3	7	60.5
N (non-risk reviews)	1,960	589.74	1,012.09	29	139	1,731

Panel B: H2 Allowance-Informativeness Sample (four-quarter horizon)

Variable	N	Mean	SD	p10	p50	p90
Future NCO, t+1–t+4 (%)	970	0.56	0.81	0.02	0.24	2.00
Allowance (%)	970	1.52	1.19	0.27	1.29	2.86
Prior-four-quarter WLB gap	970	0.24	0.51	-0.34	0.24	0.84
Current-quarter NCO (%)	970	0.13	0.21	0.00	0.06	0.46
Current change in NPL (%)	970	0.01	0.17	-0.12	0.00	0.14
Log(Assets)	970	18.76	1.49	16.62	18.93	20.88
Capital ratio	970	0.11	0.03	0.08	0.10	0.15
Loan growth (%)	970	1.43	4.33	-2.45	0.89	5.41
EBLLP ratio	970	0.03	0.03	0.01	0.02	0.06
Deposit ratio	970	0.65	0.18	0.38	0.73	0.83

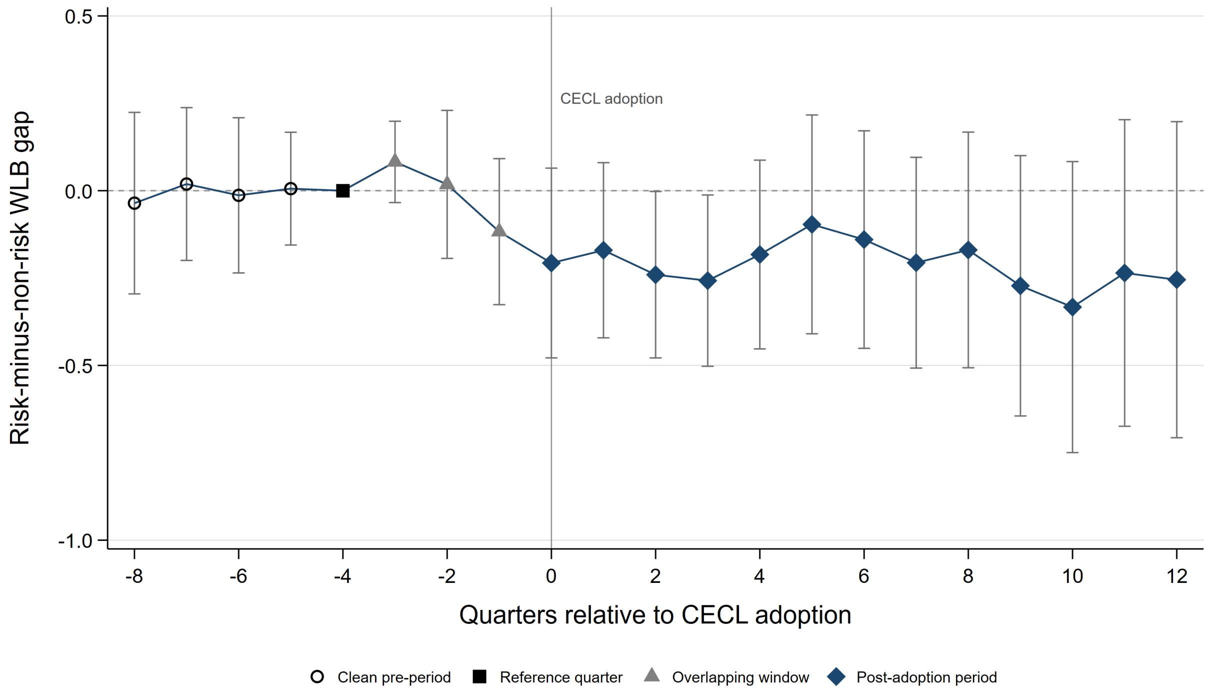
Panel A reports the *H1* estimation sample. Panel B reports the four-quarter H2 same-bank sample, comprising 970 bank-quarters from 45 BHCs represented in both accounting regimes after singleton removal. *WLB_Gap* equals mean risk-function WLB minus mean non-risk WLB. Each group must have at least three total reviews and an observed WLB mean; group means use nonmissing ratings. Reviews without job titles enter the non-risk group. Panel A pools reviews posted in quarters t through $t+3$; Panel B pools quarters $t-4$ through $t-1$. *Post* indicates CECL adoption. Future NCO sums net charge-offs over $t+1$ through $t+4$. The allowance and credit-loss measures are scaled by gross loans. Financial measures come from FR Y-9C filings and use the definitions and transformations in the corresponding regressions. Variables marked (%) are displayed in percentage units.

Table 2: **CECL Adoption and Employee Work-Life Balance**

	Panel A: Full H1 sample			**Panel B: Late adopters**		
	(1) WLB Gap	(2) Risk WLB	(3) Non-risk WLB	(4) WLB Gap	(5) Risk WLB	(6) Non-risk WLB
Post	-0.168**	-0.168*	0.000	-0.606***	-0.543**	0.063
	(0.082)	(0.091)	(0.040)	(0.211)	(0.216)	(0.061)
Log(Assets)	-0.342**	-0.351**	-0.009	-0.568*	-0.331	0.238
	(0.172)	(0.159)	(0.126)	(0.285)	(0.247)	(0.180)
Capital ratio	-1.949	-2.247	-0.298	-2.819	-1.608	1.211
	(1.850)	(2.427)	(1.533)	(3.338)	(3.772)	(1.977)
EBLLP ratio	-2.132*	-2.013*	0.119	-3.081**	-2.649	0.432
	(1.138)	(1.199)	(0.571)	(1.318)	(1.888)	(1.079)
Deposit ratio	0.852	0.956	0.104	-2.902*	-1.985	0.918
	(0.737)	(0.846)	(0.356)	(1.619)	(1.907)	(0.623)
Observations	1,960	1,960	1,960	378	378	378
BHCs	128	128	128	35	35	35
Within R^2	0.0203	0.0224	0.0007	0.0964	0.0581	0.0503
BHC FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes

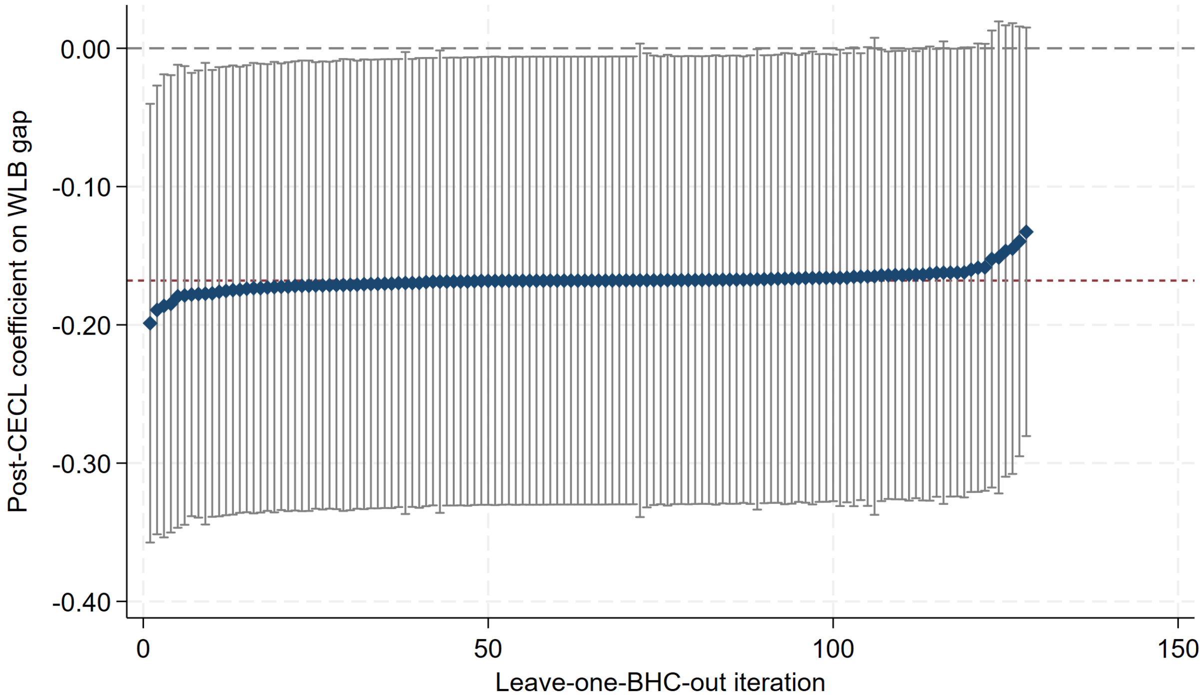
The dependent variables are the WLB gap and its risk and non-risk components. Each group must have at least three total reviews over quarters t through $t+3$ and an observed WLB mean. Panel A uses the full *H1* sample. Panel B includes BHCs adopting in 2022Q1 or later and never-treated controls. *Post* equals one from the adoption quarter onward. The three regressions within each panel use identical observations and include BHC and quarter fixed effects. The Panel B gap estimate has a wild-bootstrap p-value of 0.0246 using 9,999 replications and Webb weights. Standard errors clustered by BHC are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Fig. 1. **Dynamic Effects of CECL Adoption on the WLB Gap**



This figure plots event-study estimates of the effect of CECL adoption on WLB_Gap, obtained by replacing $Post_{i,t}$ in Equation (2) with indicators for event quarters $k = -8$ through $k = 12$ relative to each BHC's adoption quarter. The specification includes the bank-level controls, BHC fixed effects, and calendar-quarter fixed effects of Equation (2). Because the review window for quarter t pools reviews posted through quarter $t + 3$, the last measurement drawing exclusively on pre-adoption reviews is dated four quarters before adoption. The reference period is therefore $k = -4$ (solid square). Hollow circles mark the clean pre-period ($k = -8$ through $k = -5$), grey triangles mark the overlapping window ($k = -3$ through $k = -1$), in which measurements partially pool post-adoption reviews, and solid diamonds mark the post-adoption period. Vertical bars are 95 percent confidence intervals based on standard errors clustered by BHC.

Fig. 2. **Leave-One-Out Distribution of the Baseline Estimate**



This figure plots the coefficient on *Post* from Equation (2), re-estimated 128 times with one BHC omitted per iteration, sorted by estimate magnitude. Vertical bars are 95 percent confidence intervals based on standard errors clustered by BHC. The horizontal line marks the full-sample estimate of -0.168 from column (1) of Table 2. Estimates range from -0.199 to -0.133. All 128 estimates are negative and significant at the 10 percent level, and 88 percent are significant at the 5 percent level.

Table 3: **CECL Adoption and Alternative Employee Rating Gaps**

	Dependent variable: risk-minus-non-risk rating gap					
	(1) WLB	(2) Overall	(3) Career	(4) Compensation	(5) Culture	(6) Leadership
Post	-0.168**	-0.027	-0.019	-0.079	0.032	-0.030
	(0.082)	(0.065)	(0.097)	(0.076)	(0.088)	(0.108)
Observations	1,960	1,960	1,960	1,960	1,960	1,960
BHCs	128	128	128	128	128	128
Within R^2	0.0203	0.0042	0.0159	0.0127	0.0056	0.0061
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes
BHC FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Average placebo coefficient	−0.024					
WLB − average placebo	−0.144*					
p-value of difference	0.094					

Each column reports the coefficient on *Post* for the indicated risk-minus-non-risk rating gap. The common sample satisfies Table 2's review-count requirement and has observed group means for all six dimensions; individual reviews need not report all six ratings. Controls and fixed effects match Table 2. The bottom rows compare WLB with the average of the other five outcomes using a stacked regression with outcome-specific coefficients, controls, and fixed effects. Standard errors clustered by BHC are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 4: **The Nature of Workforce Strain Following CECL Adoption**

	Coefficient on Post		
Complaints as DV	**(1) Gap**	**(2) Risk**	**(3) Non-risk**
Workload	0.040**	0.035**	-0.005
	(0.017)	(0.017)	(0.006)
Workflow Frictions	-0.032	-0.030	0.002
	(0.030)	(0.027)	(0.009)
Role Pressure	-0.020	-0.024	-0.004
	(0.017)	(0.016)	(0.006)
Job Security	-0.007	-0.012	-0.005
	(0.022)	(0.025)	(0.009)
Autonomy	-0.019	-0.018	0.001
	(0.016)	(0.015)	(0.007)
Managerial Support	0.053**	0.044*	-0.009
	(0.023)	(0.022)	(0.011)
Career Rewards	0.069**	0.035	-0.033***
	(0.033)	(0.032)	(0.009)
Organizational Justice	-0.005	-0.013	-0.008*
	(0.018)	(0.017)	(0.005)
Observations	1,966	1,966	1,966
BHCs	128	128	128
Bank controls	Yes	Yes	Yes
BHC FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes

Each cell reports the coefficient on *Post* from a separate regression of the indicated complaint rate or risk-minus-non-risk rate gap. Rates are proportions of classified *Cons* reviews containing the category; categories can overlap. Reviews are pooled over quarters t through $t+3$. The common sample independently requires at least three classified *Cons* reviews per group and available measures for all eight categories, without requiring WLB ratings or membership in the *H1* sample. Controls, BHC fixed effects, and quarter fixed effects match Table 2. Standard errors clustered by BHC are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 5: **Workforce Conditions and Allowance Informativeness across Future-Loss Horizons**

Future NCO horizon	(1) t+1	(2) t+1 to t+2	(3) t+1 to t+4
Panel A: All eligible banks			
Pre: Allowance × WLB_Gap	-0.0013	-0.0002	0.0078
	(0.0038)	(0.0066)	(0.0107)
Post: Allowance × WLB_Gap	0.0049	0.0303***	0.0895***
	(0.0064)	(0.0111)	(0.0219)
Post − Pre	0.0063	0.0305**	0.0818***
	(0.0090)	(0.0142)	(0.0252)
Observations	1,446	1,440	1,430
BHCs	110	109	108
Within R^2	0.3507	0.3365	0.2803
Panel B: Same-bank sample			
Pre: Allowance × WLB_Gap	-0.0015	-0.0007	0.0062
	(0.0035)	(0.0064)	(0.0101)
Post: Allowance × WLB_Gap	0.0050	0.0357***	0.0904***
	(0.0105)	(0.0081)	(0.0311)
Post − Pre	0.0065	0.0363***	0.0842**
	(0.0135)	(0.0113)	(0.0362)
Observations	976	974	970
BHCs	45	45	45
Within R^2	0.4152	0.4020	0.3558
Controls	Yes	Yes	Yes
BHC × regime FE	Yes	Yes	Yes
Quarter × regime FE	Yes	Yes	Yes

The dependent variable is cumulative net charge-offs over the indicated future horizon, divided by quarter-t gross loans. *Allowance* is the quarter-t allowance divided by gross loans. *WLB_Gap* uses reviews pooled over $t-4$ through $t-1$, with at least three total reviews per group and both WLB means observed. These four consecutive quarters and quarter t must lie within one accounting regime. Both panels include CECL adopters. Panel B additionally requires observations in both regimes after singleton removal at each horizon. Allowance, WLB, their interaction, controls, and BHC and quarter fixed effects are fully interacted with accounting regime. Controls include size, capitalization, loan growth, EBLLP, deposits, current NCO, and the change in NPL. Only the interaction coefficients and their Post − Pre differences are displayed. Focal observations end in 2024Q4; 2025 filings supply future NCO only. Standard errors clustered by BHC are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 6: **Risk- versus Non-risk-Function Workforce Conditions**

	All eligible banks		Same-bank sample	
	$t+1$ to $t+2$	$t+1$ to $t+4$	$t+1$ to $t+2$	$t+1$ to $t+4$
Risk WLB: Post − Pre	0.0310*	0.0871***	0.0388**	0.0915***
	(0.0157)	(0.0189)	(0.0177)	(0.0257)
Non-risk WLB: Post − Pre	-0.0455	-0.0725	-0.1034*	-0.2339*
	(0.0634)	(0.1628)	(0.0525)	(0.1346)
Observations	1,440	1,430	974	970
BHCs	109	108	45	45
Controls	Yes	Yes	Yes	Yes
BHC × regime FE	Yes	Yes	Yes	Yes
Quarter × regime FE	Yes	Yes	Yes	Yes

Risk-function and non-risk-function WLB enter jointly with their allowance interactions and lower-order terms. The table reports the Post − Pre change in each allowance interaction. Outcomes, review requirements, samples, controls, and regime-interacted BHC and quarter fixed effects match Table 5. Standard errors clustered by BHC are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 7: **Inference and Influence Diagnostics: Same-Bank Sample**

Future NCO horizon	(1) $t+1$ to $t+2$	(2) $t+1$ to $t+4$
Allowance × WLB_Gap		
Post	0.0357	0.0904
Clustered p-value	< 0.0001	0.0057
Wild-bootstrap p-value	0.0005	0.0985
Leave-one-BHC-out range	[0.0241, 0.0443]	[0.0599, 0.1253]
LOO estimates significant at 10%	100.0%	97.8%
Post − Pre	0.0363	0.0842
Clustered p-value	0.0024	0.0249
Wild-bootstrap p-value	0.0025	0.2000
Leave-one-BHC-out range	[0.0252, 0.0697]	[0.0540, 0.1286]
LOO estimates significant at 10%	100.0%	97.8%

The same-bank sample contains 45 BHCs represented in both regimes after singleton removal, with 974 observations at the two-quarter horizon and 970 at the four-quarter horizon. Specifications match Tables 5. Wild-bootstrap p-values use 9,999 replications, Webb weights, and BHC clustering. Leave-one-BHC-out diagnostics omit each of the 45 banks in turn and report coefficient ranges and the percentage significant at 10% using conventional BHC-clustered inference.

Table 8: **Job-Title Availability and WLB Measurement**

Panel A: WLB Measures on the Common H1 Estimation Sample

Common sample: 1,955 bank-quarters

	Mean		Median		SD		Pearson r
	All	Titled-only	All	Titled-only	All	Titled-only	
Risk WLB	3.7646	3.7646	3.8148	3.8148	0.6880	0.6880	1.0000
Non-risk WLB	3.5085	3.5096	3.5284	3.5328	0.3771	0.3840	0.9621
WLB Gap	0.2561	0.2550	0.2678	0.2693	0.6270	0.6349	0.9862

Panel B: Job-Title Availability by Review-Posting Year, 2015–2025

Review year	Total reviews	With title	Without title	Without title (%)
2015	24,218	13,115	11,103	45.85
2016	25,434	13,073	12,361	48.60
2017	23,109	11,054	12,055	52.17
2018	19,540	8,645	10,895	55.76
2019	19,959	15,203	4,756	23.83
2020	25,458	23,064	2,394	9.40
2021	57,695	52,102	5,593	9.69
2022	54,478	47,419	7,059	12.96
2023	47,038	44,405	2,633	5.60
2024	49,325	46,369	2,956	5.99
2025	33,146	30,435	2,711	8.18
Total	**379,400**	**304,884**	**74,516**	**19.64**

All retains reviews without reported job titles in the non-risk group; Titled-only excludes them before aggregation. Panel A uses the intersection of the two *H1* estimation samples, with equal weight per bank-quarter and the review requirements in Table 2. Each correlation compares the two measures on these same observations. Panel B counts source review records posted from 2015 through 2025, before bank-mapping and regression-sample restrictions. Its percentages describe source coverage rather than regression-sample composition.

Table 9: **Robustness to Excluding Reviews without Reported Job Titles**

Panel A: CECL Adoption and Employee Work-Life Balance

Sample	WLB Gap	Risk WLB	Non-risk WLB	Observations	BHCs
Baseline	-0.156*	-0.167*	-0.011	1,955	128
	(0.083)	(0.091)	(0.041)		
Late adopters	-0.590***	-0.543**	0.047	378	35
	(0.214)	(0.216)	(0.069)		

Panel B: The Nature of Workforce Strain

Complaint category	Gap	Risk	Non-risk	Observations	BHCs
Workload	0.034*	0.035**	0.001	1,961	128
	(0.019)	(0.017)	(0.007)		
Managerial Support	0.052**	0.044*	-0.008	1,961	128
	(0.024)	(0.022)	(0.011)		

Panel C: Allowance Informativeness across Future-Loss Horizons

	All eligible banks			Same-bank sample		
Future NCO horizon	t+1	t+1 to t+2	t+1 to t+4	t+1	t+1 to t+2	t+1 to t+4
Pre: Allowance × WLB_Gap	-0.0006	0.0015	0.0092	-0.0010	0.0009	0.0074
	(0.0035)	(0.0062)	(0.0078)	(0.0034)	(0.0061)	(0.0078)
Post: Allowance × WLB_Gap	0.0057	0.0313***	0.0913***	0.0056	0.0359***	0.0889***
	(0.0062)	(0.0112)	(0.0219)	(0.0097)	(0.0078)	(0.0311)
Post − Pre	0.0063	0.0298**	0.0820***	0.0067	0.0349***	0.0815**
	(0.0086)	(0.0137)	(0.0240)	(0.0126)	(0.0098)	(0.0339)
Observations	1,441	1,435	1,425	973	971	967
BHCs	110	109	108	45	45	45

Reviews without reported job titles are excluded before aggregation. Panels A, B, and C repeat Tables 2, 4, and 5, respectively, retaining their windows, controls, and fixed effects. Panels A and C require at least three total reviews per group and observed group WLB means. Panel B instead requires at least three classified *Cons* reviews per group and available measures for all eight categories, independently of the WLB sample; only workload and managerial support are displayed. Within each row of Panels A and B, gap and component regressions use identical observations. Panel C's same-bank sample requires both regimes after singleton removal at each horizon. Standard errors clustered by BHC are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Appendix A. Variable Definitions

Table A1: **Variable Definitions**

Variable	Definition	Source
CECL adoption		
Post	Indicator equal to one in and after the identified adoption quarter and zero otherwise. Adoption quarter is the first quarter in which BHC i reports a nonmissing value for BHCKJJ26, BHCKJJ27, or BHCKJJ28.	FR Y-9C
Employee reviews and ratings		
WLB	Work/Life Balance rating, ranging from one to five, with higher values indicating better work-life balance.	Glassdoor
WLB_Gap	Risk-function mean WLB minus non-risk mean WLB for the same BHC and review window. A higher gap indicates better relative WLB in the risk function.	Glassdoor
Other rating gaps	Risk minus non-risk mean ratings for Overall, Career Opportunities, Compensation and Benefits, Culture and Values, and Senior Management. Each rating ranges from one to five.	Glassdoor
Textual measures		
Complaint	Share of successfully classified *Cons* reviews assigned to a complaint category within a function, BHC, and forward review window.	Glassdoor
Complaint gap	Risk-function complaint incidence minus non-risk incidence for the same category, BHC, and window. Both incidence and gaps are measured as proportions; 0.01 represents one percentage point.	Glassdoor
Allowance and realized credit losses		
Allowance	Quarter-end allowance for loan and lease losses (BHCK3123), divided by quarter-end gross loans and leases (BHCK2122).	FR Y-9C

Continued on next page

Table A1 continued

Variable	Definition	Source
Future NCO, h quarters	$\sum_{s=1}^{h} NCO_{i,t+s}^{\text{amt}}/L_{i,t}$, where $h \in \{1,2,4\}$, $L_{i,t}$ is quarter-t gross loans, and NCO^{amt} is quarterly gross charge-offs (BHCK4635) minus recoveries (BHCK4605). Year-to-date charge-offs and recoveries are converted to quarterly flows before summation.	FR Y-9C
Current NCO	$NCO_{i,t}^{\text{amt}}/[(L_{i,t}+L_{i,t-1})/2]$. The denominator is the average of current and previous quarter-end gross loans.	FR Y-9C
Change in NPL	$(NPL_{i,t} - NPL_{i,t-1})/L_{i,t-1}$. The NPL series uses BHCK5526 before 2018 and BHCK1403 from 2018 onward.	FR Y-9C
Bank controls		
Size / Log(Assets)	Natural logarithm of total consolidated assets (BHCK2170), using the reporting units in the regulatory data.	FR Y-9C
Capital ratio	Total equity capital including noncontrolling interests (BHCKG105), divided by total assets.	FR Y-9C
EBLLP ratio	Income before taxes plus provision expense (BHCK4301 + BHCK4230), divided by previous quarter-end gross loans. The numerator uses the reported amounts without converting them to quarterly flows.	FR Y-9C
Deposit ratio	Sum of interest-bearing and noninterest-bearing deposits in domestic and foreign offices (BHDM6631, BHDM6636, BHFN6631, and BHFN6636), divided by total assets. Missing components contribute zero to the sum.	FR Y-9C
Loan growth	$(L_{i,t} - L_{i,t-1})/L_{i,t-1}$, where L is gross loans and leases.	FR Y-9C
Log(Employees)	Natural logarithm of reported full-time-equivalent employees (BHCK4150).	FR Y-9C

Financial ratios enter regressions in decimal units and are multiplied by 100 where table headings indicate percentages. Size, the capital, EBLLP, and deposit ratios, loan growth, the allowance ratio, current and future NCO ratios, and the change in NPL are winsorized at the 1st and 99th percentiles. Employee ratings, complaint measures, and their gaps are not winsorized.

Appendix B. Supplementary Analyses

Table B1: **Pre-Adoption WLB Determinants and Gap Validation**

	Pre-adoption risk-function WLB			*Pre-adoption WLB_Gap*	
	(1)	(2)	(3)	(4)	(5)
Log(Assets)	−0.043	−0.220	−0.075	−0.018	−0.058
	(0.054)	(0.227)	(0.203)	(0.041)	(0.199)
Capital ratio		−4.700	−2.727		−2.500
		(3.668)	(2.857)		(2.880)
EBLLP ratio		1.888	0.992		0.888
		(3.587)	(3.407)		(3.473)
Deposit ratio		0.079	0.204		0.219
		(0.624)	(0.473)		(0.471)
Log(Employees)		0.181	0.063		0.050
		(0.206)	(0.182)		(0.183)
Pre-adoption non-risk WLB			0.897***		
			(0.261)		
BHCs	67	67	67	67	67
Adjusted R^2	−0.0046	−0.0164	0.2397	−0.0126	−0.0515

This table reports BHC-level OLS regressions for 67 adopters. Risk and non-risk WLB are averaged over eligible quarters dated before adoption, with weights equal to the total review count for the corresponding function. The gap is the difference between these two averages. Each quarterly WLB measure pools reviews from t through $t + 3$ and can therefore include reviews posted after adoption. Financial controls and Log(Employees) are averaged over the same quarters. All columns use the same sample. Variables are defined in Appendix A. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table B2: **Staggered-Adoption Estimates of the Effect of CECL Adoption on the WLB Gap**

	Estimate	Std. error	p-value	95% CI
ATT, adoption-quarter measurement	−0.348*	(0.181)	0.055	[−0.703, 0.007]
Average ATT, event quarters 0 through +4	−0.287**	(0.129)	0.026	[−0.539, −0.035]
Observations			801	
BHCs			128	

This table reports Callaway and Sant'Anna (2021) estimates with not-yet-treated banks as the comparison group. The adoption date used in estimation is shifted three quarters earlier because WLB pools reviews from t through $t+3$. This excludes windows containing post-adoption reviews from the untreated comparison group. Reported event times are relative to actual adoption. The first row covers the first four post-adoption quarters; the second averages estimates for event quarters zero through four. Estimation uses the regression-based method without covariates. Standard errors are clustered by BHC. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table B3: **Workforce Conditions and Allowance Informativeness: Control Ladder**

Specification	(1) Literature	(2) + Non-credit	(3) + Current NCO	(4) + Change in NPL
Panel A: All eligible banks				
Pre: Allowance × WLB_Gap	0.0053	0.0085	0.0079	0.0078
	(0.0128)	(0.0108)	(0.0109)	(0.0107)
Post: Allowance × WLB_Gap	0.0843***	0.0846***	0.0875***	0.0895***
	(0.0259)	(0.0252)	(0.0227)	(0.0219)
Post − Pre	0.0790***	0.0761***	0.0796***	0.0818***
	(0.0280)	(0.0264)	(0.0261)	(0.0252)
Observations	1,430	1,430	1,430	1,430
BHCs	108	108	108	108
Within R^2	0.1400	0.1637	0.2703	0.2803
Panel B: Same-bank sample				
Pre: Allowance × WLB_Gap	0.0041	0.0071	0.0066	0.0062
	(0.0123)	(0.0103)	(0.0105)	(0.0101)
Post: Allowance × WLB_Gap	0.1090***	0.1076***	0.0896***	0.0904***
	(0.0363)	(0.0321)	(0.0315)	(0.0311)
Post − Pre	0.1049**	0.1005***	0.0830**	0.0842**
	(0.0415)	(0.0360)	(0.0370)	(0.0362)
Observations	970	970	970	970
BHCs	45	45	45	45
Within R^2	0.1385	0.1742	0.3452	0.3558
Size, capital, and loan growth	Yes	Yes	Yes	Yes
EBLLP and deposit ratios	No	Yes	Yes	Yes
Current NCO	No	No	Yes	Yes
Current change in NPL	No	No	No	Yes
BHC × regime FE	Yes	Yes	Yes	Yes
Quarter × regime FE	Yes	Yes	Yes	Yes

This table reports the four-quarter specification in Table 5 with successively larger control sets. The dependent variable is cumulative NCO over $t+1$ through $t+4$, divided by gross loans at t. Allowance is measured at t and WLB over $t-4$ through $t-1$. Within each panel, the sample from column (4) is held fixed across columns. Both panels include adopters; Panel B retains BHCs represented in both regimes after singleton removal. All regressors and fixed effects are interacted with accounting regime. Only the Allowance × WLB_Gap coefficients and their differences across regimes are shown. Standard errors clustered by BHC are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Appendix C. Glassdoor Data Construction

C.1. Bank Matching

Glassdoor reviews may appear under the name of a BHC or one of its subsidiaries. I use the WRDS Bank Structure database to identify commercial-bank subsidiaries and search Glassdoor using both parent and subsidiary names.

I construct two search variants of each name. The first removes legal and geographic suffixes; the second also removes terms such as "Bank," "Bancorp," and "Trust," while retaining at least four characters. I preserve periods in abbreviations and append "Bank" to single-word stems. Searches proceed from the first variant to the original name and then the second variant, returning up to three candidate pages per query.

I compare candidate names using Levenshtein, token-sort, and token-set similarity scores. A match must satisfy three conditions. The first substantive word must match, allowing a common prefix of at least four characters; the candidate name must be no more than twice as long as the query; and the highest similarity score must be at least 95 on a 0–100 scale. Queries of five characters or fewer require a Levenshtein score of at least 95. For entries that remain unmatched, I conduct a second search using the original names and additional variants that adjust punctuation or remove regional qualifiers, parenthetical text, or the appended "Bank" suffix. I examine up to ten candidate pages per query. The minimum similarity score is 92, increasing to 94 for the aggressive variant and 95 for variants that remove regional qualifiers. Queries of five characters or fewer still require a character-based similarity score of at least 95. The second round retains restrictions on first-word compatibility and token overlap, while lowering the minimum ratio of query length to candidate-name length to 0.40.

C.2. Review Extraction and Processing

I collect the available reviews from each matched page, including posting dates, ratings, job titles, employment status, location, and written comments. I check that each downloaded file can be read and contains reviews. Reviews are assigned to calendar quarters by posting date; records without a usable date are excluded from the quarterly measures. These measures use reviews posted from 2015Q1 onward, including later reviews needed to complete forward windows near the end of the focal period. Table D2 uses the full review history collected.

C.3. Function Classification and Aggregation

The classifier operates in two stages. In the first stage, I normalize each self-reported job title by lowercasing, collapsing whitespace, standardizing punctuation, folding unicode dash variants, and stripping pure seniority modifiers that do not carry role-defining meaning[7]. Role-defining nouns that function as the head of the title are preserved[8]. The output of this stage is a base role for each review (e.g., risk manager, credit analyst, operations director). Working from base roles rather than raw job titles reduces classification noise from minor title variants.

In the second stage, the resulting base role is matched to a predefined classification dictionary, whose assignments take precedence over the keyword rules. For titles absent from the dictionary, I check regular-expression patterns for accounting first, followed by risk and economics, and then information technology. Titles that match none of these patterns and reviews without reported titles are assigned to the other category.

[7]The stripped tokens include senior, sr, jr, junior, principal, staff, lead, assistant, associate, intern, temporary, contract, contractor, global, regional, corporate, and numeric level suffixes (I, II, III, IV, V, level 1, level 2, and so on).

[8]For example, manager, director, vice president (and its variants VP, AVP, SVP, EVP), chief, head of, and supervisor.

Appendix D. Text Classification

I use a large language model to classify *Cons* comments into the eight categories in Table D1. A review may be assigned to more than one category if it describes multiple complaints. The model receives the review text and a record identifier, but no separate information on the employer, job function, review date, ratings, or CECL adoption. All reviews are classified using the same rules.

The model assigns a zero or one to each category and separately identifies reviews containing no substantive complaint. I classify a review as *Other Complaints* if all eight indicators equal zero and the review is not classified as *No Complaint*. I check that the returned labels follow these rules. Reviews without usable *Cons* text remain unclassified. Reviews assigned to either residual category are included in the denominator of the complaint shares.

Table D2 reports the frequency of each category. Table D3 relates the complaint indicators to ratings from the same reviews. Panel B excludes comments that explicitly mention work-life balance to assess whether the association between workload complaints and WLB extends beyond that wording. Section D.1 reproduces the classification rules.

Table D1: **Complaint Categories and Illustrative Reviews**

Complaint category	Coding rule	Key distinction	Illustrative review excerpt
Workload	Excessive work amount or intensity, time demands, long hours, deadlines, understaffing or capacity constraints, burnout, or explicit negative work-life balance.	Generic pressure, stress, targets, or performance demands alone are not sufficient unless excessive work, time, or capacity demands are described.	"Sometimes too much work to be done in short time."
Autonomy	Restricted employee discretion or autonomy over how, when, or where work is performed, including micromanagement or lack of decision authority.	Hierarchy, rules, or governance alone are not sufficient unless the employee's own discretion is restricted.	"Your time is micromanaged from your breaks to your lunch."
Workflow Frictions	Workflow, process, system, data, documentation, coordination, or approval problems that make work harder to complete, transmit, review, or validate.	Vague inefficiency, bureaucracy, or poor management alone is not sufficient without an identifiable obstacle to completing the work.	"Lots of red tape and hard to get things done."
Role Pressure	Unclear, conflicting, changing, or unusually demanding expectations, targets, scrutiny, accountability, or responsibility.	Generic stress or pressure alone is not sufficient; the source of the demand must be identifiable from the review.	"Difficult goals that change constantly and almost impossible to reach."
Managerial Support	Lack of managerial or coworker support, hostility, disrespect, poor leadership relationships, abusive behavior, or other failures of interpersonal support.	Micromanagement by itself is classified as Autonomy; identifiable process or system problems are classified as Workflow Frictions.	"Unorganized management, no guidance or leadership."
Job Security	Layoff or job-loss risk, organizational instability, restructuring uncertainty, or insecurity about continued employment.	Turnover or organizational change alone is not sufficient unless the review describes instability, uncertainty, or employment insecurity.	"Not knowing if you will have your job the next week."
Career Rewards	Complaints about compensation, benefits, promotion, recognition, career advancement, training, or professional development.	Training problems that interfere with completing current work may instead be classified as Workflow Frictions; training problems concerning development or advancement are classified here.	"Promotion is limited and can take years."
Organizational Justice	Unfair treatment, favoritism, discrimination, inconsistent treatment, procedural unfairness, or inequitable organizational decisions.	Politics or general dissatisfaction alone is not sufficient unless the review describes unfair, biased, preferential, or inconsistent treatment.	"There is somewhat of a culture of favoritism."
Residual categories			
No Complaint	The *CONS* text contains no meaningful negative complaint.	This category includes responses such as "none," "no cons," positive comments, or other text containing no substantive criticism.	—
Other Complaints	The review contains a meaningful negative complaint, but none of the eight complaint categories above applies.	This category is constructed mechanically when all eight complaint categories are coded zero and the review is not classified as *No Complaint*.	—

The eight complaint categories are coded independently, and a review may receive more than one label. Examples are excerpts from the review data. *No Complaint* and *Other Complaints* are mutually exclusive residual categories. Section D.1 provides the classification rules.

Table D2: **Distribution of Complaint Categories**

Complaint category	All reviews (%)	Risk function (%)	Non-risk function (%)
Workload	20.94	16.58	21.09
Autonomy	11.32	10.48	11.35
Workflow Frictions	16.76	17.44	16.74
Role Pressure	13.37	9.90	13.49
Managerial Support	21.42	19.03	21.50
Job Security	8.08	9.29	8.04
Career Rewards	35.81	39.91	35.67
Organizational Justice	9.60	8.46	9.64
No Complaint	8.93	7.95	8.96
Other Complaints	10.80	11.23	10.78

This table reports category frequencies for 430,215 classified reviews, including 14,459 risk and 415,756 non-risk reviews. It uses the full review history collected, including reviews outside the focal period. Each review record is counted once, even when its employer page is matched to more than one BHC. The eight complaint categories can overlap, so their percentages need not sum to 100%. Reviews classified as *No Complaint* or *Other Complaints* receive no substantive category label.

Table D3: **Textual Complaint Dimensions and Employee Ratings**

Complaint dimension	(1) WLB	(2) Overall	(3) Career	(4) Compensation	(5) Culture	(6) Leadership
Panel A: Risk-function reviews						
Workload	**-0.935***	-0.281***	-0.114***	-0.158***	-0.246***	-0.213***
	(0.039)	(0.031)	(0.032)	(0.029)	(0.032)	(0.040)
Autonomy	-0.226***	-0.246***	-0.246***	-0.082**	-0.231***	-0.273***
	(0.042)	(0.034)	(0.032)	(0.036)	(0.037)	(0.029)
Workflow Frictions	-0.055*	-0.216***	-0.178***	-0.135***	-0.152***	-0.215***
	(0.028)	(0.028)	(0.040)	(0.033)	(0.032)	(0.033)
Role Pressure	-0.144***	-0.208***	-0.175***	-0.055*	-0.193***	-0.288***
	(0.044)	(0.034)	(0.040)	(0.033)	(0.042)	(0.042)
Managerial Support	**-0.750***	**-1.043***	**-0.845***	**-0.532***	**-1.157***	**-1.233***
	(0.035)	(0.034)	(0.043)	(0.038)	(0.037)	(0.035)
Job Security	-0.240***	-0.409***	-0.425***	-0.076**	-0.400***	-0.443***
	(0.043)	(0.038)	(0.039)	(0.032)	(0.046)	(0.035)
Career Rewards	0.007	-0.314***	**-0.510***	**-0.613***	-0.093***	-0.259***
	(0.033)	(0.026)	(0.020)	(0.039)	(0.023)	(0.023)
Organizational Justice	-0.388***	**-0.616***	**-0.584***	-0.281***	**-0.667***	**-0.672***
	(0.044)	(0.034)	(0.041)	(0.045)	(0.039)	(0.038)
Reviews	9,750	9,750	9,750	9,750	9,750	9,750
Glassdoor pages	197	197	197	197	197	197
Glassdoor-page FE	Yes	Yes	Yes	Yes	Yes	Yes
Review-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Excluding direct work-life-balance wording						
Workload	**-0.775***	-0.215***	-0.093**	-0.149***	-0.180***	-0.131**
	(0.041)	(0.033)	(0.039)	(0.033)	(0.039)	(0.051)
Reviews	9,428	9,428	9,428	9,428	9,428	9,428
Glassdoor pages	197	197	197	197	197	197
Glassdoor-page FE	Yes	Yes	Yes	Yes	Yes	Yes
Review-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes

Panel A reports review-level OLS regressions for risk-function reviews on matched Glassdoor pages from 2015Q1 onward. Each column includes all eight complaint indicators, Glassdoor-page fixed effects, and review-quarter fixed effects. The six columns use the same reviews with all ratings and classifications available. Each review is counted once, including those on pages matched to multiple BHCs. Panel B excludes comments containing common variants of "work-life balance" or the standalone abbreviation "WLB"; only the workload coefficient is reported. Standard errors clustered by Glassdoor page are in parentheses. Coefficients of at least 0.50 in absolute value are shown in bold. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

D.1. Operational Classification Definitions

The following definitions are used in the classification prompt. Examples and instructions for formatting the output are omitted. Labels must reflect what the review states or clearly implies. Negative tone alone is insufficient, and the classifier is instructed not to infer consequences that the review does not describe. Each category is assessed independently.

1. Workload

Core question. Does the review describe excessive work demands, excessive time demands, lack of work-life balance, insufficient capacity, or excessive work intensity?

Code 1 for. Excessive workload or work volume; too many tasks or responsibilities; long hours, overtime, nights, or weekends caused by work; understaffing or insufficient capacity that raises work demands; unrealistic or excessive deadlines; excessive work intensity or overwork; burnout explicitly tied to workload, hours, staffing, or work intensity; and explicit negative work-life-balance complaints.

Do not code 1 for. Positive statements that work-life balance is good; generic stress or pressure with no work-demand or work-life-balance content; performance pressure by itself; scrutiny or accountability by itself; difficult judgment by itself; inefficient processes by themselves; or micromanagement by itself.

2. Autonomy

Core question. Does the employee lack meaningful control over how, when, or where work is performed?

Code 1 for. Micromanagement; lack of autonomy or discretion; inability to make work-related decisions; rigid rules that directly constrain employee judgment; lack of control over schedule; inflexible hours or vacation arrangements; forced office presence or lack of location flexibility when framed as a loss of flexibility or control; inability to plan one's work because imposed priorities remove discretion; and responsibility without corresponding authority.

Do not code 1 for. A rude, abusive, or incompetent manager without an autonomy issue; slow workflows or approvals without a personal-control issue; unclear expectations by themselves; excessive workload by itself; or bureaucracy, policies, and hierarchy merely because they exist. These conditions qualify only when the review indicates that they restrict the employee's discretion over how, when, or where work is performed.

3. Workflow Frictions

Core question. Does the organizational production environment create an identifiable obstacle that makes work unnecessarily difficult to complete, coordinate, transmit, review, or approve?

Code 1 for. Bureaucracy or red tape that obstructs completing work; excessive approval layers that delay or complicate work; inefficient processes or workflows; manual processes or reconciliation; outdated, fragmented, or inadequate systems or technology; data-quality, data-availability, or information-access problems; organizational silos; coordination failures across teams or functions; poor operational handoffs; duplicated work or rework; unclear ownership when it obstructs task completion; documentation or administrative burden; and operational communication failures that impede getting work done.

Do not code 1 for. Being slow to change by itself; poor change management without an identifiable operational obstacle; generic poor leadership communication; generic interpersonal conflict; workload without a process or coordination obstacle; unclear performance expectations without a workflow obstacle; or rigid supervision that primarily reflects an autonomy issue.

4. Role Pressure

Core question. Does the employee face unclear, conflicting, changing, high-stakes, unusually scrutinized, or explicitly demanding work expectations?

Code 1 for. Unclear expectations; role ambiguity; conflicting responsibilities; competing priorities; constantly changing priorities or moving targets; explicit performance pressure or aggressive targets; excessive scrutiny or review; pressure to defend decisions or assumptions; heightened responsibility or accountability for outcomes; high-stakes judgment; fear of mistakes when tied to responsibility or accountability; incompatible demands; and repeated urgency arising from competing or changing expectations.

Do not code 1 for. Excessive work volume or hours when that is the only complaint; lack of autonomy when expectations themselves are clear; inefficient systems or processes alone; or generic stress, pressure, a high-pressure environment, or a demanding job without an identifiable source in targets, expectations, scrutiny, accountability, ambiguity, conflict, or changing priorities.

5. Managerial Support

Core question. Does the employee describe a lack of supportive, respectful, competent, or functional managerial or interpersonal relationships?

Code 1 for. Unsupportive management; poor leadership when the complaint concerns people or support; hostile or toxic managers; bullying; abusive or disrespectful treatment; managers who do not listen; lack of coaching, help, or support; dysfunctional team relationships; hostile coworkers; poor interpersonal communication; and social isolation or lack of collegial support.

Do not code 1 for. Micromanagement alone; favoritism, discrimination, or unfairness without an independent support problem; operational handoff or silo problems; or changing or unclear expectations without an independent managerial or interpersonal support issue.

6. Job Security

Core question. Does the review describe meaningful instability in the organization or employment relationship that makes the employee's work or future less secure or predictable?

Code 1 for. Layoffs or fear of layoffs; job insecurity; repeated restructuring or reorganizations explicitly creating instability; merger or acquisition disruption; department closure or role elimination; turnover explicitly described as creating instability or uncertainty; uncertainty about continued employment; and organizational instability affecting employees.

Do not code 1 for. New leadership or organizational change by itself; turnover mentioned only because it creates understaffing; ordinary organizational change without instability, disruption, or employment uncertainty; changing daily priorities; or lack of promotion opportunity.

7. Career Rewards

Core question. Is the complaint about economic rewards, advancement, recognition, development, or long-term career opportunity?

Code 1 for. Low pay or salary; poor bonuses; inadequate benefits; lack of promotion; weak career paths or advancement; inadequate recognition; limited professional development; lack of training when the complaint concerns career development; and poor growth opportunities.

Do not code 1 for. Fear of losing one's job; employment insecurity; or training deficiencies that primarily prevent completion of current work rather than limit career development.

8. Organizational Justice

Core question. Does the employee explicitly describe unfair, inconsistent, discriminatory, preferential, retaliatory, biased, or arbitrary treatment?

Code 1 for. Favoritism; nepotism; discrimination; unequal or inconsistent treatment; unfair promotion or pay decisions; arbitrary treatment; retaliation; biased decision making; organizational politics when the review clearly describes politics as unfair allocation, preferential treatment, or biased decision making; and unfair or opaque procedures when fairness itself is the complaint.

Do not code 1 for. Office politics or politics alone; generic ethics concerns; generic diversity concerns; dishonesty alone; low pay alone; lack of promotion opportunity alone; rude management without a fairness issue; rigid rules applied consistently to everyone; or inefficient or opaque workflows when fairness itself is not the complaint.